

SIMULATION OF A CAR WASH QUEUE

You are tasked with designing a simulation of a car wash queue system for a car wash facility. The facility has multiple washing stations, and cars arrive at the facility for washing throughout the day.

The car wash operates as follows:

Cars arrive at the facility and join a queue to wait for their turn to be washed.

Each washing station can handle one car at a time. When a washing station becomes available, the next car in the queue is moved to the station for washing.

Once a car has been washed, it leaves the station, and the next car in the queue moves up for washing.

Your task is to implement a data structure and algorithm to simulate this car wash queue system efficiently. The solution should support the following operations:

Enqueue: Add a new car to the end of the queue.

Dequeue: Remove the next car from the front of the queue.

Check if the queue is empty.

Display the current state of the queue (optional).

Students should analyze the problem requirements and consider which data structure (linked list, stack, or queue) would be most suitable for efficiently implementing the car wash queue system. They can then design and implement the solution using the chosen data structure.

Input:

1. Enqueue: Car1
2. Enqueue: Car2
3. Enqueue: Car3
4. Dequeue
5. Enqueue: Car4
6. Dequeue
7. Check if queue is empty
8. Display current state of queue

Output:

1. Car1 added to the queue.
2. Car2 added to the queue.
3. Car3 added to the queue.
4. Car1 removed from the queue and sent for washing.
5. Car4 added to the queue.
6. Car2 removed from the queue and sent for washing.
7. Queue is not empty.
8. Current state of the queue:
Car3 (Next in line for washing), Car4

